

PAUL CHUKWU

Embedded Systems and Edge AI Developer | Systems Integration | Hardware Validation
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Professional Summary

Computer Engineering student specializing in **embedded systems**, **edge AI**, and **systems integration** with experience designing, validating, and deploying intelligent hardware solutions. Hands-on experience developing AI applications on **NVIDIA Jetson Orin Nano** and **Raspberry Pi** platforms, integrating radar and vision sensors, optimizing embedded performance, and documenting engineering results. Strong foundation in **C/C++**, **Python**, **FPGA/RTL** design, **Linux**, hardware validation, communication protocols, and collaborative **Agile** development.

Engineering Experience

Computational Analysis and Acceleration Research Group (CARG), uOttawa

2026 to Present

Embedded AI Developer, Volunteer

Ottawa, ON

- Reduced **Jetson Orin Nano** AI inference latency by over **50%** through model balancing and deployment profiling for a Human Activity Recognition system.
- Integrated **TI radar** point cloud and **Novelda** heart/breath sensors into **Linux**-based embedded AI pipelines while validating sensor interfaces, hardware compatibility, and model inputs.
- Developed benchmarking procedures for multiple AI models, evaluating latency, runtime stability, **GPU** utilization, and deployment feasibility on embedded hardware.
- Documented benchmark results, deployment configurations, and engineering findings for biweekly Chief Engineer reviews, supporting reproducible validation and technical handoff.

Qinvst

Sep 2024 to Dec 2024

Software Validation Intern

Ottawa, ON

- Designed regression-style test cases for UI/**API** workflows, validating inputs, boundary cases, error handling, data consistency, and integration paths before sprint delivery.
- Reproduced and triaged defects in **Jira** with test steps, expected vs. actual behavior, screenshots/logs, and validation notes for developer handoff.
- Collaborated in **Agile** sprint cycles using **Git** and **Jira**, participating in testing reviews, issue prioritization, validation reporting, and release readiness.

Snappy Techs

Aug 2021 to Jan 2023

Computer Technician Intern

Ottawa, ON

- Diagnosed, repaired, and validated laptops/desktops through hardware fault isolation, OS setup, storage health checks, peripheral testing, repair documentation, and post-service stability checks.
- Troubleshoot **Windows** operating systems, storage devices, memory modules, and peripheral hardware while maintaining detailed repair and validation records.

Technical Skills

Programming: C, C++, Python, Java, SQL, Bash

Embedded Systems: Jetson Orin Nano, Raspberry Pi 4B/5, Arduino, PlatformIO, GPIO, UART, I2C, SPI, interrupts, cameras, sensors, servos, OLED/LCD, oscilloscopes, multimeters, soldering

AI and Data: YOLOv8, OpenCV, ONNX, model deployment, benchmarking, **TI radar**, **Novelda radar**, point clouds, pose estimation, object detection, MySQL, PostgreSQL

Systems and Networking: Linux, Windows 11, VMware, TCP/IP, OSI Model, REST APIs, client-server architecture, hardware/software integration, technical documentation

FPGA and Verification: SystemVerilog, Verilog, VHDL, FSM/ASM design, RTL debug, testbenches, waveform debugging, directed tests, regression testing, Quartus-II Pro, ModelSim

Development Tools: Git, GitHub, Jira, VS Code, CLion, Arduino IDE, PlatformIO

Education

University of Ottawa | BAsC in Computer Engineering | Ottawa, ON

Expected June 2026

Coursework: Microcontrollers, FPGA Design, Digital Systems, Computer Architecture, Operating Systems, Circuit Design, Data Structures, C/C++. Activities: IEEE Vice Chair, CEGSC Academic Events Team. Awards: Dean's List, Honor Roll, UOSU Scholarship for Innovation in Sustainability.

Selected Embedded and Hardware Projects

- UO Supermileage Steering Wheel and Telemetry System** | *C/C++*, *Sensors*, *Interrupts*: Developed embedded vehicle telemetry and steering wheel features using magnetic sensing, speedometer logic, timing calculations, interrupt-driven inputs, and terminal/display output.
- Ecollector / Esotrash Self Recycling Machine** | *Raspberry Pi 5*, *C++*, *PlatformIO*, *MySQL/PostgreSQL*: Developed an AI-powered smart recycling system using camera vision, ultrasonic sensors, servo motors, and **C++** control logic; logged waste classifications into **MySQL/PostgreSQL** databases and achieved 97.5% validation accuracy.
- Assistive Smart Glasses for Blind Users** | *YOLOv8*, *Raspberry Pi 4B/5*: Created wearable object and obstacle detection system using camera vision; improved embedded performance by migrating deployment from **Raspberry Pi 4B** to **Raspberry Pi 5** after latency testing.
- UART Traffic Light Controller on FPGA** | *VHDL*, *Verilog*, *Quartus-II Pro*, *ModelSim*: Designed structural HDL FSM logic with **UART** transmission, baud timing, counters, comparators, debouncers, and waveform-based verification of 8-bit serial data flow.
- Ross Video FPGA Display System** | *SystemVerilog*, *Quartus-II Pro*: Developed a 1080x720 FPGA display system using coordinate-based rendering and hardware timing constraints; earned 2nd Place at the Hack the Hill FPGA Challenge.

Leadership and Activities

IEEE Vice Chair, University of Ottawa Student Branch | CEGSC Academic Events Team | Volunteer Researcher, CARG

Awards and Achievements

2nd Place, Hack the Hill Ross Video FPGA Challenge | 1st Place, Eco Sorter Self Recycling Bin | Six Sigma Certified | CPR/AED and First Aid Certified